

PROJECT II

RELATIONSHIP BETWEEN THE CANADA-US EXCHANGE RATE AND COMMODITY PRICES

II

RELATIONSHIP BETWEEN THE CANADA-US EXCHANGE RATE AND COMMODITY PRICES

Extended Analysis · Structural Breaks · Forecasting

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Introduction

Owing to globalization, especially regarding trade, the relationship between exchange rates and commodity prices has attracted attention in the literature. Since this helps in forecasting better, either reduce losses or avoid them. Ferraro and Rossi (2015), from their findings on the relationship between exchange rate and commodity prices, found that having known future prices of commodities, the exchange rate can be forecasted. Since their paper focused on oil prices, they concluded that with a good model of forecasting oil prices, it is possible to predict the exchange rates in the future. Also, Chen et al. (2008), from their regression between the two variables, found that the exchange rates could predict commodity prices strongly; however, the opposite was weak. Which is, the commodity prices were weak with forecasting the exchange rates. This is supported by Kohlscheen et al. (2017), who found that the link between commodity prices and exchange rates is statistically significant at a high frequency. Also, Bashar and Kabir (2013), using cointegration and error correction models from using data from the third quarter of 1982 to the second quarter of 2013, found that the coefficients of interest rate and commodity prices in Australia were positive and significant on the exchange rate of the country in the long run. Similarly, for the Canada-US exchange rate and commodity prices, Choudhri and Schembri (2014), through empirical analysis, found that there is a positive and significant long-run effect of productivity and commodity prices indices on the exchange rate. Meanwhile, this has shifted and has a more robust and positive trend since 1990. Furthermore, from the empirical results of Jung et al. (2020), both short-run and long-run asymmetric relationships existed between the Canada-US exchange rate and the real oil price. In addition, Mow (2013) saw that commodity prices explain variations in the US-Canadian dollar exchange rate and other currencies. These emphasize support the relationship

between commodity prices and the exchange rate as is the focus of this paper, especially with the Canada-US exchange rates and commodity prices.

Theoretical Framework

Milijakovic (1999) presents that the exchange rate is referred to in simple language as the local or home-currency price of a foreign currency, that is, quoting the foreign currency in the face of the home currency. And the commodity prices are the set prices for commodities and items as determined by the forces of demand and supply in the market. As such, the relationship that exists between the two variables, commodity prices and the exchange rate, can be discussed in light of two theories being the Present-Value approach and the “Law on One Price”.

In line with Chen et al. (2010) is a theory expounding the relationship that exists between the commodity prices and the exchange rates, which is the Present-Value Approach. According to them, the present value method presents that the exchange rate should Granger-cause its fundamentals which is equivalent to either the consequence of endogenous response or the outcome of reverse causality. In addition, monetary policy responses to exchange rate leads to a Granger-cause money supply or interest rate. Furthermore, the exchange rate changes may occur before inflation when prices are sticky or hardly changes. In such instances, Chen et al. (2010) present that the positive Granger-cause will be challenging to explain the present value framework except for instances where the fundamental is exogenous and not endogenous to the exchange rate. In effect, the commodity prices are exogenous and a unique fundamental to the exchange rate. Therefore, with the present-value approach, the exchange rate can predict the world commodity prices when even the commodity prices do not forecast exchange rates in the future. This is because the exchange rate harbours information about the commodity prices in the future; meanwhile, the commodity prices cannot do the same since they cannot predict their future costs. Therefore, conclude that there is a less direct linkage between the two variables.

Another analysis of the relationship between these two variables stems from the “law of one price”. The Law of one price states that once prices are converted to a common currency, the same good should sell for the same price in different countries”, Milijakovic

(1999, pg.126). According to Isard, P. (1977), the law of one price prevails when there are no transport costs trade barriers. The perfect commodity arbitrage gives way for uniform pricing of goods in common currency units worldwide. Therefore, if a particular good is valued in a currency for trade common to different economies, the price or value should not differ within these different economies with the same currency, but they all should have a uniform price – “one price” across. Notwithstanding,

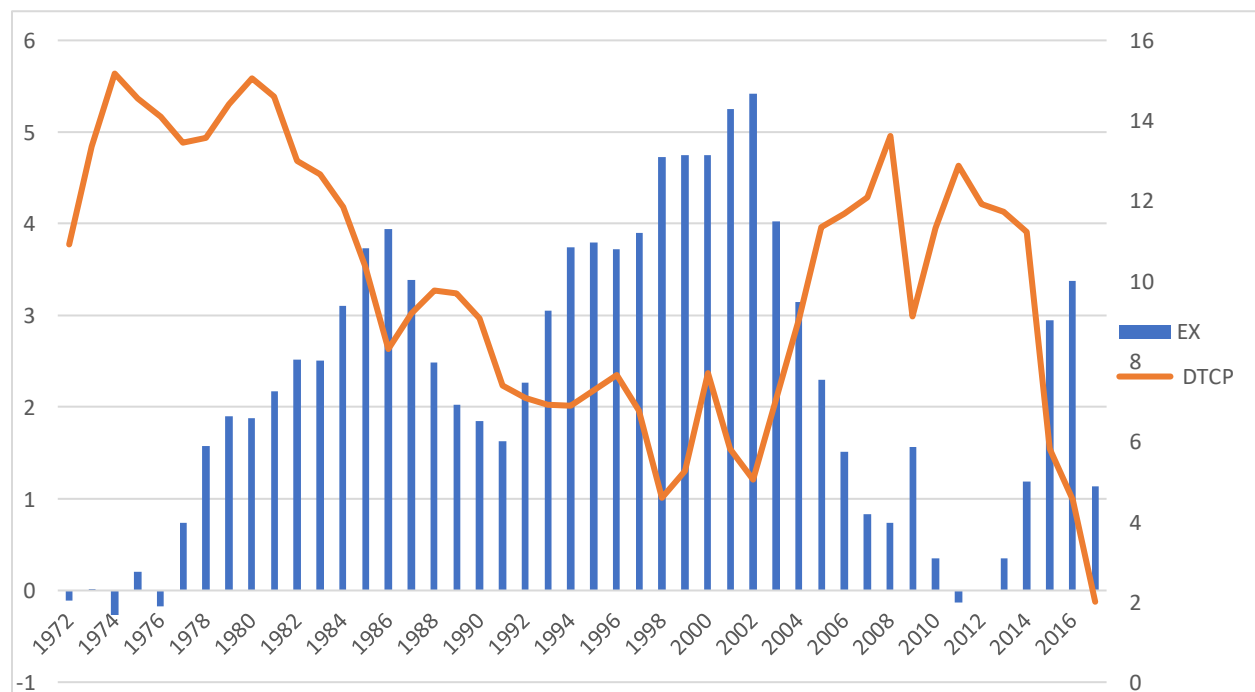
there are arguments in the literature that the “law of one price” is violated [the likes of Richardson, 1978; Isard 1977]. However, Rogoff found that the Law of one price applies better with countries that engage with highly traded commodities, for example, gold. Therefore, this law is referred to as an essential ingredient in the field of international economics and trade whose absence implies no complete theory of international trade, as opined by Officer (1986).

Meanwhile, the Canadian dollar, which is the focus of this paper, according to Laidler and Aba (2001), appreciated against some currencies and depreciated against others from the findings of their report from 1973-2000. The paper concluded that the Canadian Dollar is a commodity currency as its value depends on the changes in commodity prices. Hence, this paper explores the Canadian dollar, a commodity currency, and its relationship with the US dollar in commodities and exchange rates.

Description of Data

This study used the same data sets as Project I. Commodity price indices and the Canada-US nominal exchange rate is among the data, which can be found on the Statistics Canada website. Because there was no recent data for the Canada-US nominal exchange rate, the study used data spanning from January 1972 to April 2017. However, for the purposes of this research, data from January 1972 to April 2017 will be used. The US commodity price index was used to deflate the total commodity prices. In addition, I transformed the deflated total commodity prices and nominal exchange rates by taking logs of both variables. As a result, the log-transformed variables were used throughout this project's analysis to transform and stabilize a series' variance.

The graph of the log of the Canada – US exchange rate and log of Deflated commodity prices.



The deflated total commodity price rises when the exchange rate is low, as shown in the graph above. The deflated total commodity prices are greatly affected by increased high-frequency variability of the Canada-US exchange rate.

Empirical Section

Unit Root and Cointegration

Unit Root Testing

In the use of time series, most macroeconomic variables are likely to be non-stationary. As a result, the variance of the process would be infinite instead of being constant, and regression with non-stationary series is expected to generate spurious regression not suitable for making correct inferences and generalizations. Therefore, stationary in time series analysis is very useful because it ensures that the results are valid without the persistence of shocks that would lead to a spurious regression. Hence, before performing any analysis, a research analyst must ensure that the data is stationary or trend stationary. In order to do test the stationarity of deflated total commodity price and Canada-US Nominal Exchange Rate, we must conduct a unit root test. For the purposes of this study, the data appears to have a trend. There is some possibility of autocorrelation; I will perform the Augmented Dickey-Fuller Test, assuming there is a trend to test for stationarity. Again, in order to avoid spurious regression that will end up affecting inference, I will use the Augmented Dickey-Fuller (ADF) test to verify whether all variables used in this study are in the order of integrated $I(1)$ or $I(0)$ because $I(1)$ variables processes can lead to spurious results that can lead to misleading inference and generalization.

The number of lags must be considered when performing the ADF test for unit root testing. The choice of lag number is essential because, while a lag that is too small can bias the test due to the remaining serial correlation in the error terms, a too-large lag can reduce the test's power to reject the null hypothesis when it is false. The optimal lag order and significance level were determined throughout the analysis using the Akaike Information Criterion and a 5% significance level, respectively. The ADF is compatible with or without trend at both levels and first difference.

The ADF is illustrated in the below equation as:

$$\Delta Y_t = \beta_0 + \beta_1 t + \beta Y_{t-1} + \sum_{j=1}^{p-1} Q_j \Delta Y_{t-j} + \epsilon_t$$

Hypothesis

$H_0: \beta = 0$ {There is a Unit root and implies that data is non – Stationary}

$H_1: \beta < 0$ {No Unit root and implies that data is Trend Stationary}

The above equation Δ denotes the first difference in the deflated nominal exchange rate and price level, Y_t is the time series being tested, t is the time trend variable, and k is the number of lags that are added to the model to ensure that the residuals, ϵ_t are white noise. The failure to rejection of the null hypothesis implies that the time series is non-stationary, whereas the rejection of the null hypothesis indicates that the time series is stationary.

Unit Root Test for the log of Canada-US Nominal Spot Exchange Rate

First, I looked for the unit root for the log of the Canada-US nominal spot exchange rate. The Augmented Dickey-Fuller (ADF) was used test to determine if there is a unit root in the data and by doing this, Stata was used to find the optimal lag to run the test using the Akaike Information Criterion (AIC). Gonzalo and Pitarakis (2002) provided the rationale for choosing AIC, demonstrating that unless the sample size is unrealistically small, AIC is by far the best performance criterion in Lag Length Estimation. Prior to the ADF test, the Lag order was determined. The AIC yields the optimal lag of 2 (See Appendix 1), I set the lag to 2 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic is -1.790 and the 5% critical

value is -3.410 . The output (see Appendix 2) shows that the absolute value 1.790 of the test statistic is smaller than the 5% critical value (absolute) of 3.410 . Therefore, at a 5% level of significance, there is enough evidence to fail to reject the null hypothesis, which means we can confidently conclude that log of the Canada-US Nominal Spot Exchange rate has a unit root and hence, non – stationary.

Because we know that the log of the Canada-US Nominal Spot Exchange rate contains a unit root, we can determine whether the first difference has a unit root or not, because if a unit root exists, we can use the first difference operator to make it stationary. In reference to Appendix 3, the optimal AIC lag for the first-difference variable is 4. As a result, to test the unit root process, we can run the Augmented Dickey-Fuller Test with four lags on the first difference. The test demonstrates (See Appendix 4) that the absolute value of the test statistic of 9.105 is greater than the critical value at a 5% significance level of 3.410 , allowing us to reject the null hypothesis that a unit root exists. As a result, the results imply that there exists $I(1)$, an integrated process of order one, in the log of the Canada-US nominal spot exchange rate.

Unit Root Test for the log of Deflated Total Commodity Price

Similarly, I check to see if the log of Deflated Total Commodity price reflects has a unit root. I determined the lag order and included it in the ADF test. The AIC yields the optimal lag of 3 (See Appendix 5) therefore I set the lag to 3 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic is 2.512 (Absolute value), and the critical value is 3.410 (5% level of significant, absolute value), we fail to reject the null hypothesis and conclude that the process has a unit root because the critical value exceeds the test statistic (See Appendix 6).

Using the first difference process, we determine that the optimal lag is 1 (See Appendix 7). The test (See Appendix 7) shows that the absolute value of the test statistic is greater than the critical value at the 5% significance level, so the null hypothesis can be rejected. The results imply that there is an $I(1)$, integrated of order one process in the log of the Deflated Price of Total of All Commodities.

Theory of Cointegration

Two variables are said to be cointegrated if their linear combination is integrated of order 0. When two variables share the same permanent shock, cointegration occurs. As a result, even if there are unit root processes in the data of two independent variables, the regression of one variable on the other will no longer be spurious. The regressions would still be spurious if the variables were not cointegrated. Cointegration tests identify scenarios in which two or more non-stationary time series are integrated together in such a way that they cannot deviate from equilibrium over time. The cointegration test in this study is used to determine whether there is a long-term relationship between two variables, namely the deflated total commodity price and the Canada-US Nominal Spot Exchange Rate. According to the theory of the present-value method, the exchange rate can predict commodity prices in the future. As a result, the cointegration test in this study determines whether our data agree with or refute the present-value assertion. The Vector Autoregressive model was then used to determine whether the variables have a short-run relationship.

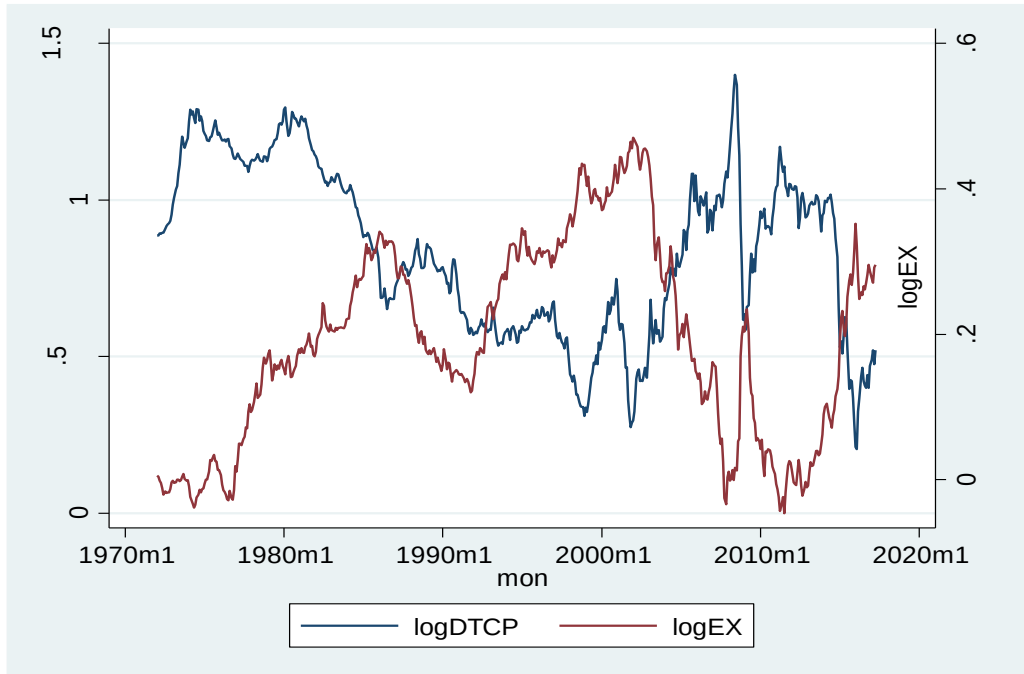
To investigate the causal relationship between deflated commodity prices and exchange rates, the Granger causality test was used. The null hypothesis for the Granger Causality Test is that the log of the Canada-US Spot Exchange Rate lagged variable does not cause or has no explaining power in predicting the log of the Deflated Price of All

Commodities and vice versa. If exchange rates granger-cause commodity prices or vice versa, the granger causality test will either confirm or reject the present value theory that exchange rates have predictive ability for forecasting deflated total commodity prices or vice versa.

I perform diagnostic testing and stability analysis to ensure that the models are correctly specified. The Jarque-Bera test, Kurtosis, and Skewness tests are among the diagnostic and stability tests. These tests look for normally distributed disturbances in the residuals, as well as autocorrelations, structural breaks, and model stability. The presence of autocorrelation in the residuals indicates that our OLS regression is inconsistent, and our estimates will be unbiased and inefficient.

Test for Cointegration

The graph below depicts the graphical relationship between the Canada-US spot exchange rate and total commodity price from 1972 to 2017. According to the diagram below, the two diagrams move together, and they exhibit similar behaviors, implying that these variables are not likely to be cointegrated.



In order to test for cointegration, I conducted the test using Stata and it shows that the AIC has an optimal lag is 2 (see Appendix 8) and I set the lag to 2 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic in ADF showed a value of -3.183 and the asymptotic critical value of 2.89 at a 5% level of significance. Hence, we fail to reject the null hypothesis and conclude that there is a unit root and therefore, the log of the Canada-US spot exchange rate and the log of the deflated total commodity are not cointegrated.

The Johansen test for cointegration was used to validate the ADF test results. An optimal lag order of 2 was chosen (See Appendix 10) and used in the Johansen test for cointegration. The restricted trend model was used throughout the analysis because it is based on the assumption that trends do not enter the cointegration space. The unrestricted trend model, on the other hand, yielded no significant result (See Appendix

11). In relation to AIC, an optimal lag of 2 was used to perform the Johansen test for cointegration, and the results are shown below.

. vecrank logDTCP logEX, trend(rtrend) lags(2)					
Johansen tests for cointegration					
Trend: rtrend		Number of obs =		542	
Sample: 1972m3 - 2017m4		Lags =		2	
				5%	
maximum			trace	critical	
rank	parms	LL	eigenvalue	statistic	value
0	6	2636.3886	.	17.4989*	25.32
1	10	2643.3389	0.02532	3.5983	12.25
2	12	2645.138	0.00662		

Statistical analysis from the table above:

Hypothesis: H_0 : There is no long-run relationship between the log of deflated commodity prices and exchange rate. (No cointegration between deflated commodity prices and exchange rate) / Rank = 0.

H_1 : There is a long-run relationship between the log of deflated commodity prices and exchange rate (deflated commodity prices and exchange rate are cointegrated).

We cannot reject the null hypothesis because our trace statistic (17.4989) is less than a 5% critical value (25.32). As a result, the Johansen Test of Cointegration confirms that the log of the Canada-US Spot Exchange Rate and the log of the deflated Price of All

Commodities are not cointegrated. This implies that the log of deflated commodity prices and the log of the Canadian-US exchange rate has no long-run relationship. Because the log of deflated commodity prices and the log of the Canadian-US exchange rate are not cointegrated, the vector error correction model will not be appropriate, rather than Vector Autoregression, is used. The unrestricted Vector Autoregression (VAR) model will suggest whether there is any short-run causality between the log of the Canada-US Spot Exchange Rate and the log of the deflated Price of All Commodities in this case.

Vector Autoregression Model (VAR)

The vector autoregression (VAR) model is one of the most widely used multivariate regression time-series analytic techniques. The VAR model is useful because it can explain past and causal relationships among multiple variables over time, as well as predict future observations. Explanation and prediction of future observations in a time series are dependent on correctly postulating a VAR model and estimating its parameters (Lutkepohl, 2005). Based on the results of the Johansen Test of Cointegration, we determined that there is no long-run causality between the two variables. The first difference in variables will be used to estimate the Vector Autoregression (VAR) model to determine whether there is any short-run causality between the log of Canada-US Spot Exchange Rate and the log of deflated Price of All Commodities. I then go on to determine the optimal lag order to run the first difference in variable and then add one extra lag number to overfit the model and have a good fit for generalization and inference (See Appendix 12). At first lag orders (a reference to Appendix 13), the chi-square values of (123.3418, 199.1056) are statistically significant and we can conclude that there is a short-run relationship between deflated commodity

prices and the Canadian-US exchange rates, and this implies that in the short term, exchange rates have an impact on commodity prices.

Diagnostic Tests for verifying the VAR model

The LM test for residual autocorrelation determines whether autocorrelation exists in the residuals for a given lag order. In general, the residuals from a VAR model should be white noise with no autocorrelation. If there is autocorrelation among the residuals, it indicates that some information was not accounted for by the model, such as insufficient lags. LM test for residual autocorrelation demonstrates that the VAR in the levels model is correctly specified. The test is based on the concept of Lagrange multiplier testing, and it is also known as an LM test for serial correlation.

The hypothesis is stated below:

H_0 There is no autocorrelation at lag order

H_1 There is autocorrelation at lag order

The following is the output from Stata:

```
. varlmar, mlag (10)
```

Lagrange-multiplier test

lag	chi2	df	Prob > chi2
1	3.1528	4	0.53258
2	7.7001	4	0.10320
3	4.7062	4	0.31879
4	7.9174	4	0.09465
5	5.7365	4	0.21971
6	7.9356	4	0.09396
7	7.9807	4	0.09229
8	2.0738	4	0.72218
9	13.4297	4	0.00936
10	5.6740	4	0.22485

H0: no autocorrelation at lag order

Because the p-value is greater than the significant level, there is insufficient evidence to reject the null hypothesis. As a result, we can conclude that no autocorrelation exists in the residuals for any lag order. This demonstrates that the VAR in the levels model is correctly specified.

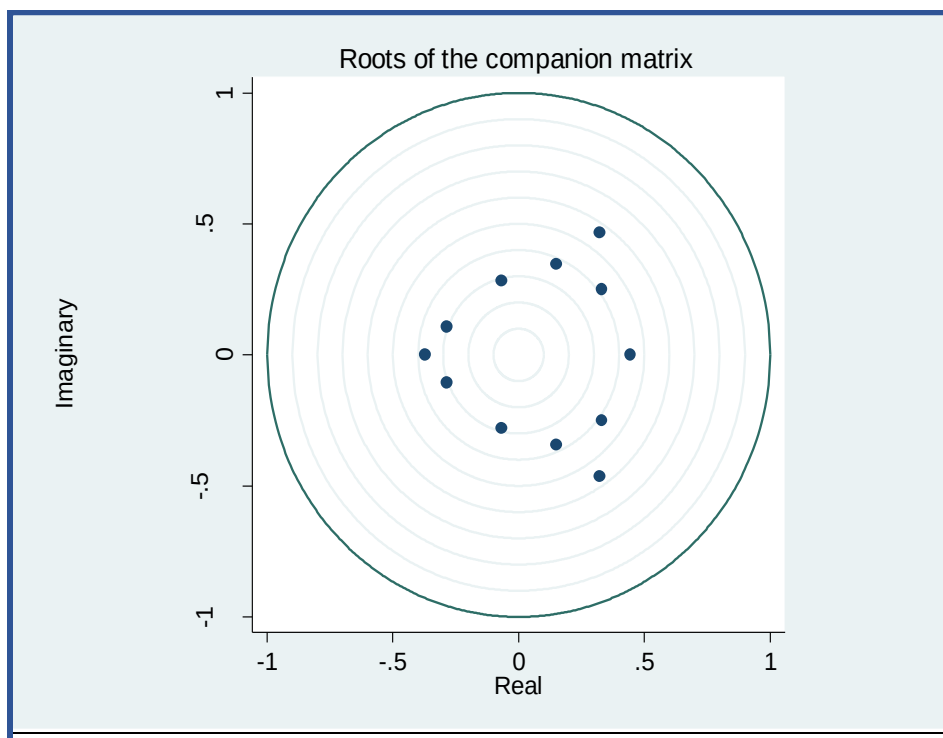
Normality Test

The assumption of normality is that the underlying residuals are normally distributed, or approximately so. The normality of the residuals is an assumption made when running a linear model. So, if your residuals are normal, your assumption is correct, and your model inference (confidence intervals, model predictions) should be correct as well. If your residuals are not normal, there could be a problem with the model's fit, stability, or reliability. To generalize a regression model beyond the sample, some of the assumptions of regression residuals must be tested. The Jarque - Beta test will be used

to determine the residuals' normality, skewness, and kurtosis. The test's null hypothesis states that residuals are normally distributed, while the alternate hypothesis states that residuals are not normally distributed. The test results in an appendix (14) indicate that at 5% significance level, we can reject the null of normality in residuals. The Skewness and Kurtosis test results are similar to the Jarque-Bera test, there is enough evidence to reject the null hypothesis. This result indicates that residuals are not normally distributed, and this implies that the VAR model may have been misspecified.

VAR Stability Test

The roots of the characteristic polynomial with the log of deflated total commodity price and log of Canada-US Nominal Exchange Rate were used to assess the stability of the VAR model. However, the eigenvalues and the unit circle would be used to assess the stability. The figure below depicts the evaluation of stability using the characteristic roots of log of deflated total commodity price and log of Canada-US Nominal Exchange Rate.



Eigenvalue stability condition

Eigenvalue	Modulus
.4067034	.406703
.09837096 + .2787719i	.295619
.09837096 - .2787719i	.295619
-.1030333	.103033

All the eigenvalues lie inside the unit circle.
VAR satisfies stability condition.

By evaluating the roots of the above VAR system, the stability of the VAR is visually inspected. The dots represent the roots of the common polynomial. If no roots are found outside the unit circle, the VAR system meets the stability condition, indicating

model stability. Because all of the roots are within the unit circle, the VAR model meets the stability conditions.

Test for Structural Breaks

A structural break occurs when a time series abruptly changes at a specific point in time. This change could be a change in the mean or the other parameters of the process that produces the series. Structural break tests assist us in determining when and if there is a significant change in our data. The null hypothesis of the test states that there is no structural break. The alternative hypothesis states that there is a structural break. I verify to see if the datasets used in the analysis have any structural breaks. In reference to appendix 15, the test statistic is statistically significant at all significant values for the log of deflated commodity prices and the log of Canada-US nominal exchange rate, hence the data set has a structural break. There is evidence of a structural break in the sample from November 1978 to July 2010. However, the first difference estimation reveals no structural breaks.

Granger Causality Test

Granger causality is a causality concept derived from the idea that causes may not always occur after effects and that if one variable is the cause of another, knowing the status of the cause at an earlier point in time can help predict the effect at a later point in time (Granger, 1969; Lütkepohl 2005, p.41). The null hypothesis for the Granger Causality Test is that the log of the Canada-US Spot Exchange Rate lagged variable does not cause or has no explaining power in predicting the log of the Deflated Price of All Commodities and vice versa. If exchange rates granger-cause commodity prices or vice versa, the granger causality test will either confirm or reject the present value theory

that exchange rates have predictive ability for forecasting deflated total commodity prices or vice versa.

A granger causality test was performed on the normal variables and the results were compared to variables in the first difference. Appendix (17) contains the test results. The p-value of the test (0.004) is less than the 5% significant level, there is enough information to reject the null hypothesis and conclude that the exchange rate granger causes commodity price. Also, we fail to reject the null hypothesis at the 5% significance level for the first difference results (see Appendix 18), implying that the exchange rate does not granger-cause commodity prices when variables are first difference stationary. Because the first difference variables are stationary, I used VAR in levels to test for causality and validate the previous result from the first difference Granger-causality test. At the 5% significance level, the result (see appendix 19) confirms the previous result that the exchange rate does not granger-cause commodity prices.

PART II

In this section, I will begin by determining whether the commodity prices in each sector are stationary, and the test will be performed using the Augmented Dickey-Fuller Test. The Augmented Dickey-Fuller (ADF) will help us to verify whether each sector's commodity prices used in this study are stationary or in the order of integrated I (1). Because I (1) variables processes can lead to spurious results that can lead to misleading inference and generalization. A cointegration test will be run to see if commodity prices in each sector are affected by the same permanent shock. In other words, cointegration

analysis will determine whether the linear combination of two individuals' commodity prices is of $I(0)$. The analysis will be performed based on a 5% level of significance throughout this section. The ADF test results, as shown in Appendix 20-24, show that the p-value of all commodity prices in each sector is greater than the specified significance level (5% significant level), with the exception of the commodity price of fish, whose p-value is less than the significant level. Finally, the commodity prices of Agriculture, Metals and Minerals, Energy, and Forestry are $I(1)$ variables, while Fish is $I(0)$. First, I looked to see if there was a long-run relationship between Agricultural commodity prices and Metal & Minerals commodity prices, or if the two variables were cointegrated. The test (Appendix 25) demonstrates that the two variables are not cointegrated at rank order zero (0). This implies that there are no permanent shocks to Agricultural and Metal & Minerals commodity prices. Furthermore, the VAR in levels was used to determine whether the two commodity prices are related in the short run. The VAR in levels demonstrates that there is no short-run relationship between Agricultural and Metal & Minerals commodity prices (See appendix 25 to 26).

Second, a cointegration test was run to examine the relationship between Metal and Mineral commodity prices and energy commodity prices. The findings (See Appendix 27) indicate that the two variables are not cointegrated, implying that there is no long-term relationship between energy commodity prices and mineral commodity prices. However, I used a unit circle to assess the stability and the Jarque - Beta test to determine the normality of the residuals for Metal & Mineral commodity prices and energy commodity prices. The results (see appendix 27) show that the model meets the stability condition because all of the points are within the unit circle, and normality tests (see appendix 27) indicate that the residuals' errors are not normally distributed.

Finally, I look for long-run relationships (cointegration) in commodity prices for Agriculture, Metals and Minerals, Energy, and Forestry. Surprisingly, the results show that these commodity prices cointegrated (See appendix 28). This implies that these commodity prices have a long-run relationship and are subject to the same permanent shock. Similarly, I used a unit circle to assess the stability and the Jarque - Beta test to determine the normality of the residuals of these commodity prices. The results (see appendix 28) show that the model meets the stability condition because all of the points are within the unit circle, and normality tests (see appendix 28) indicate that their residuals' errors are not normally distributed. implying that the model was incorrectly specified. Agriculture and forestry commodity prices appear to be fine in terms of skewness, while others may be. The Kurtosis test demonstrates that all variables may have extreme values. (Appendix 28) shows the results of the autocorrelation test. We cannot reject the null hypothesis of no autocorrelation in residuals because the p-values are greater than the level of significance.

Conclusion

The empirical work on the relationship between the Canada-US Spot exchange rate and Deflated Commodity prices, and this study suggests that the Nominal Spot Exchange Rate has no predictability of Deflated Commodity Prices. When the data is first difference stationary, Exchange rates do not granger-cause commodity prices. The Johansen Test of Cointegration confirms that the Canada-US Spot Exchange Rate and the Deflated price of All Commodities are not cointegrated. This implies that Deflated commodity prices and the Canadian-US exchange rate have no long-run relationship. The Vector Autoregression model confirms that there is a short-run relationship

between deflated commodity prices and Canadian-US exchange rates, implying that exchange rates have an impact on commodity prices in the short term. I also discovered that, with the exception of fish commodity prices, all other commodity prices have a long-run relationship. The unit circle was used to evaluate the model's stability, and the Jarque - Beta test was used to determine the normality of the residuals commodity prices (except fish), and the results show that the model meets the stability condition because all of the points are within the unit circle, and normality tests show that their residuals' errors are not normally distributed. implying that the model was specified incorrectly.

APPENDIX

Appendix 1: Selecting a lag order for the log Canada-US exchange rate

```
. varsoc logEX, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	320.167				.01802	-1.1784	-1.17218	-1.1625
1	1519.3	2398.3	1	0.000	.000213	-5.61593	-5.6066	-5.59209
2	1542.75	46.9*	1	0.000	.000196*	-5.69908*	-5.68664*	-5.66729*
3	1543.23	.96385	1	0.326	.000196	-5.69716	-5.68162	-5.65742
4	1543.24	.00573	1	0.940	.000197	-5.69346	-5.67481	-5.64578

Endogenous: logEX
Exogenous: mon _cons

Appendix 2: Augmented Dickey-Fuller test of Unit root for the log Canada-US exchange rate.

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. dfuller logEX, lag (2) trend regress
```

Augmented Dickey-Fuller test for unit root

Number of obs = 541

Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-3.960	-3.410	-3.120

MacKinnon approximate p-value for Z(t) = 0.7096

D.logEX	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
logEX					
L1.	-.0080356	.0044894	-1.79	0.074	-.0168545 .0007834
LD.	.3001866	.0430225	6.98	0.000	.2156732 .3847
L2D.	-.0422755	.0431839	-0.98	0.328	-.1271058 .0425549
_trend	-2.06e-07	3.90e-06	-0.05	0.958	-7.86e-06 7.45e-06
_cons	.002005	.001384	1.45	0.148	-.0007137 .0047236

Appendix 3: Optimal lag order selection for the first difference in variables

```
. varsoc d.logEX, exo(mon)
```

Selection-order criteria
Sample: 1972m6 - 2017m4

Number of obs = 539

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	1515.03				.000213	-5.61421	-5.60799	-5.5983
1	1537.78	45.49	1	0.000	.000197	-5.6949	-5.68556*	-5.67102*
2	1538.39	1.2282	1	0.268	.000197	-5.69347	-5.68102	-5.66163
3	1538.41	.03567	1	0.850	.000198	-5.68982	-5.67426	-5.65003
4	1542.58	8.3434*	1	0.004	.000196*	-5.70159*	-5.68292	-5.65384

Endogenous: D.logEX
Exogenous: mon _cons

Appendix 4: Augmented Dickey-Fuller test for the first difference for the log Canada-US exchange rate.

```
. dfuller d.logEX, lag (4) trend regress
```

Augmented Dickey-Fuller test for unit root

Number of obs = 538

Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-3.960	-3.410	-3.120

MacKinnon approximate p-value for Z(t) = 0.0000

D2.logEX	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
D.logEX					
L1.	-.6793529	.0746141	-9.10	0.000	-.8259279
LD.	-.0196114	.0704032	-0.28	0.781	-.1579144
L2D.	-.0606174	.0617104	-0.98	0.326	-.1818439
L3D.	-.1073011	.0528136	-2.03	0.043	-.2110504
L4D.	.0241239	.0434798	0.55	0.579	-.0612896
_trend	-1.64e-06	3.87e-06	-0.42	0.672	-9.24e-06
_cons	.0008561	.0012186	0.70	0.483	-.0015377

Appendix 5: Selecting a lag order for the log Deflated Total Commodity Price

```
. varsoc logDTCP, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	1.86314				.05858	.000507	.006723	.016402
1	1010.72	2017.7	1	0.000	.001402	-3.7323	-3.72298	-3.70846
2	1034.13	46.809*	1	0.000	.00129	-3.81528	-3.80285*	-3.78349*
3	1035.15	2.0432	1	0.153	.00129*	-3.81536*	-3.79982	-3.77562
4	1035.26	.22636	1	0.634	.001294	-3.81208	-3.79343	-3.76439

Endogenous: logDTCP
Exogenous: mon _cons

Appendix 6: Augmented Dickey-Fuller test of Unit root for the log Deflated Total Commodity Price.

```
. dfuller logDTCP, lag (3) trend regress
```

Augmented Dickey-Fuller test for unit root

Number of obs = 540

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)	-2.512	-3.960	-3.410	-3.120

MacKinnon approximate p-value for Z(t) = 0.3221

D.logDTCP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
logDTCP						
L1.	-.0162617	.0064744	-2.51	0.012	-.0289801	-.0035433
LD.	.2723781	.0431299	6.32	0.000	.1876531	.3571031
L2D.	.0670255	.0447865	1.50	0.135	-.0209538	.1550049
L3D.	-.0205666	.0434653	-0.47	0.636	-.1059506	.0648173
_trend	-.0000163	.0000111	-1.47	0.142	-.0000381	5.48e-06
_cons	.0176754	.0074978	2.36	0.019	.0029466	.0324042

Appendix 7: Optimal lag order selection for the first difference in variables & Augmented Dickey-Fuller test for the first difference for the log Deflated Total Commodity Price.

varsoc d.logDTCP, exo(mon)

Selection-order criteria

Sample: 1972m6 - 2017m4

Number of obs = 539

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	1006.67				.001408	-3.72791	-3.72168	-3.71199
1	1028.73	44.114*	1	0.000	.001302*	-3.80604*	-3.7967*	-3.78216*
2	1029.4	1.3503	1	0.245	.001303	-3.80483	-3.79238	-3.773
3	1029.68	.56319	1	0.453	.001307	-3.80217	-3.7866	-3.76238
4	1030.08	.80069	1	0.371	.00131	-3.79994	-3.78127	-3.75219

Endogenous: D.logDTCP

Exogenous: mon _cons

dfuller d.logDTCP, lag(1) trend regress

Augmented Dickey-Fuller test for unit root

Number of obs = 541

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-13.205	-3.960	-3.410
MacKinnon approximate p-value for Z(t) = 0.0000			

D2.logDTCP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
D.logDTCP						
L1.	-.6832981	.0517447	-13.21	0.000	-.7849449	-.5816513
LD.	-.0502558	.0432509	-1.16	0.246	-.1352174	.0347059
_trend	-3.45e-06	9.89e-06	-0.35	0.727	-.0000229	.000016
_cons	.0004847	.0031005	0.16	0.876	-.005606	.0065753

Appendix 8: Selecting a lag order for the Cointegration test log Canada-US exchange rate and log Deflated Total Commodity Price.

Selection-order criteria								
Sample: 1972m5 - 2017m4					Number of obs		=	540
lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	573.034				.007037	-2.11864	-2.11554	-2.1107
1	1466.33	1786.6	1	0.000	.000258	-5.42343	-5.41722	-5.40754
2	1475.29	17.925*	1	0.000	.000251*	-5.45293*	-5.4436*	-5.42908*
3	1475.97	1.3704	1	0.242	.000251	-5.45176	-5.43933	-5.41997
4	1476	.05256	1	0.819	.000252	-5.44815	-5.43261	-5.40842
Endogenous: e								
Exogenous: _cons								

Appendix 9: Augmented Dickey-Fuller test for Cointegration

. dfuller e, lag (2)regress						
Augmented Dickey-Fuller test for unit root				Number of obs	=	541
	Test Statistic	----- 1% Critical Value	Interpolated Dickey-Fuller 5% Critical Value	----- 10% Critical Value		
Z(t)	-3.183	-3.430	-2.860	-2.570		
MacKinnon approximate p-value for Z(t) = 0.0210						
D.e	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
e						
L1.	-.0256315	.0080533	-3.18	0.002	-.0414513	-.0098117
LD.	.1882085	.0427957	4.40	0.000	.104141	.2722759
L2D.	-.0502585	.0429099	-1.17	0.242	-.1345505	.0340334
_cons	.0004001	.0006787	0.59	0.556	-.0009331	.0017333

Appendix 10: Optimal lag selection for Johansen test for cointegration (Restricted)

```
. varsoc logDTCP logEX, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	575.247				.000413	-2.11573	-2.1033	-2.08394
1	2593.14	4035.8	4	0.000	2.4e-07	-9.57459	-9.54973	-9.51101
2	2633.87	81.459*	4	0.000	2.1e-07*	-9.71063*	-9.67333*	-9.61526*
3	2637.5	7.2585	4	0.123	2.1e-07	-9.70925	-9.65952	-9.5821
4	2638.89	2.7891	4	0.594	2.1e-07	-9.6996	-9.63744	-9.54066

Endogenous: logDTCP logEX
Exogenous: mon _cons

Appendix 11: Optimal lag selection & Johansen test for cointegration (Unrestricted)

```
. vecrank logDTCP logEX, trend(rtrend) lags(2)
```

Johansen tests for cointegration
Trend: rtrend
Sample: 1972m3 - 2017m4

Number of obs = 542
Lags = 2

rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	6	2636.3886	.	17.4989*	25.32
1	10	2643.3389	0.02532	3.5983	12.25
2	12	2645.138	0.00662		

```
. vecrank logDTCP logEX, lags(2)
```

Johansen tests for cointegration
Trend: constant
Sample: 1972m3 - 2017m4

Number of obs = 542
Lags = 2

rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	6	2636.3886	.	12.0671*	15.41
1	9	2640.6592	0.01564	3.5259	3.76
2	10	2642.4221	0.00648		

Appendix 12: Optimal lag selection to test for VAR

```
. varsoc d.logDTCP d.logEX, exo(mon)
```

Selection-order criteria

Sample: 1972m6 - 2017m4

Number of obs = 539

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	2581.11				2.4e-07	-9.56254	-9.55009	-9.53071
1	2619.38	76.555*	4	0.000	2.1e-07*	-9.68973*	-9.66483*	-9.62606*
2	2623.06	7.3529	4	0.118	2.1e-07	-9.68853	-9.65118	-9.59303
3	2624.68	3.2504	4	0.517	2.1e-07	-9.67972	-9.62991	-9.55238
4	2629.08	8.7959	4	0.066	2.1e-07	-9.6812	-9.61894	-9.52202

Endogenous: D.logDTCP D.logEX

Exogenous: mon _cons

Appendix 13: Vector error – correction model

```
. vec d.logDTCP d.logEX, trend(rtrend) lags(2)
```

Vector error-correction model

Sample: 1972m4 - 2017m4

Number of obs = 541

AIC = -9.431119

Log likelihood = 2561.118

HQIC = -9.400083

Det(Sigma_ml) = 2.65e-07

SBIC = -9.351758

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D2_logDTCP	4	.040484	0.1871	123.3418	0.0000
D2_logEX	4	.014899	0.2709	199.1056	0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D2_logDTCP						
_cel						
L1.	-.2287421	.0484075	-4.73	0.000	-.323619	-.1338651
logDTCP						
LD2.	-.269342	.0485877	-5.54	0.000	-.3645721	-.1741119
logEX						
LD2.	.4318453	.135027	3.20	0.001	.1671971	.6964934
_cons	-8.27e-07	.0017422	-0.00	1.000	-.0034154	.0034138
D2_logEX						
_cel						
L1.	-.1944876	.0178153	-10.92	0.000	-.229405	-.1595703
logDTCP						
LD2.	.085083	.0178816	4.76	0.000	.0500357	.1201304
logEX						
LD2.	-.0119931	.0496937	-0.24	0.809	-.109391	.0854048
_cons	9.72e-07	.0006412	0.00	0.999	-.0012557	.0012576
Cointegrating equations						
Equation	Parms	chi2	P>chi2			
_cel	1	320.5451	0.0000			
Identification: beta is exactly identified						
Johansen normalization restriction imposed						
beta	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
_cel						
logDTCP						
D1.	1	.	.	.	.	.
logEX						
D1.	3.415409	.1907647	17.90	0.000	3.041517	3.789301
_trend	.0000118	.0000142	0.83	0.408	-.0000161	.0000396
_cons	-.0043613	.	.	.	.	.
.						

```
. var d.logDTCP d.logEX, lags(1/2) exog(mon)
```

Vector autoregression

Sample:	1972m4 - 2017m4	Number of obs	=	541
Log likelihood	= 2634.598	AIC	=	-9.695372
FPE	= 2.11e-07	HQIC	=	-9.658129
Det(Sigma_ml)	= 2.02e-07	SBIC	=	-9.600139

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_logDTCP	6	.035875	0.0863	51.089	0.0000
D_logEX	6	.013923	0.0928	55.37263	0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D_logDTCP						
logDTCP						
LD.	.2457558	.0471706	5.21	0.000	.153303	.3382085
L2D.	.0215443	.0473054	0.46	0.649	-.0711725	.1142611
logEX						
LD.	-.1064613	.1212633	-0.88	0.380	-.344133	.1312104
L2D.	-.1502968	.1210735	-1.24	0.214	-.3875965	.087003
mon	-4.26e-06	9.84e-06	-0.43	0.665	-.0000235	.000015
_cons	.0014163	.0043816	0.32	0.747	-.0071715	.0100041
D_logEX						
logDTCP						
LD.	-.036549	.018307	-2.00	0.046	-.07243	-.000668
L2D.	-.0146058	.0183593	-0.80	0.426	-.0505893	.0213777
logEX						
LD.	.2546562	.0470624	5.41	0.000	.1624156	.3468969
L2D.	-.0680833	.0469888	-1.45	0.147	-.1601796	.0240129
mon	-1.84e-06	3.82e-06	-0.48	0.630	-9.32e-06	5.64e-06
_cons	.0011785	.0017005	0.69	0.488	-.0021544	.0045115

Appendix 14: Normality test for log Canada-US exchange rate and log Deflated Total Commodity Price.

```
. varnorm, jbera skewness kurtosis
```

Jarque-Bera test

Equation	chi2	df	Prob > chi2
D_logDTCP	221.620	2	0.00000
D_logEX	206.693	2	0.00000
ALL	428.313	4	0.00000

Skewness test

Equation	Skewness	chi2	df	Prob > chi2
D_logDTCP	-.4614	19.195	1	0.00001
D_logEX	.23294	4.893	1	0.02697
ALL		24.088	2	0.00001

Kurtosis test

Equation	Kurtosis	chi2	df	Prob > chi2
D_logDTCP	5.9967	202.425	1	0.00000
D_logEX	5.992	201.800	1	0.00000
ALL		404.225	2	0.00000

Appendix 15: Stability analysis (structural breaks)

```
. regress logDTCP
```

Source	SS	df	MS	Number of obs	=	544
Model	0	0	.	F(0, 543)	=	0.00
Residual	39.6305562	543	.07298445	Prob > F	=	.
				R-squared	=	0.0000
				Adj R-squared	=	0.0000
Total	39.6305562	543	.07298445	Root MSE	=	.27016

logDTCP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
_cons	.8399144	.0115829	72.51	0.000	.8171617 .8626671


```
. estat sbsingle
```

	1	2	3	4	5
.....					
.....					50
.....					100
.....					150
.....					200
.....					250
.....					300
.....					350
.....					

Test for a structural break: Unknown break date

Number of obs = 544

Full sample: 1972m1 - 2017m4
Trimmed sample: 1978m11 - 2010m7
Estimated break date: 1984m11
Ho: No structural break

Test	Statistic	p-value
swald	486.7950	0.0000

Coefficients included in test: _cons

```
.
. regress logEX
```

Source	SS	df	MS	Number of obs	=	544
Model	0	0	.	F(0, 543)	=	0.00
Residual	10.067124	543	.018539823	Prob > F	=	.
				R-squared	=	0.0000
				Adj R-squared	=	0.0000
Total	10.067124	543	.018539823	Root MSE	=	.13616

logEX	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
_cons	.1908354	.0058379	32.69	0.000	.1793679 .2023029


```
. estat sbsingle
```

	1	2	3	4	5
.....					
.....					50
.....					100
.....					150
.....					200
.....					250
.....					300
.....					350
.....					

Test for a structural break: Unknown break date

Number of obs = 544

Full sample: 1972m1 - 2017m4
Trimmed sample: 1978m11 - 2010m7
Estimated break date: 1978m11
Ho: No structural break

Test	Statistic	p-value
swald	210.7065	0.0000

Coefficients included in test: _cons

Appendix 17: Granger-causality Test – first difference

. quietly var d.logDTCP d.logEX, lag(1/2)exo(mon)						
. regress logDTCP mon L(1/3).logDTCP L(1/3).logEX						
Source	SS	df	MS	Number of obs	=	541
Model	38.9491379	7	5.56416255	F(7, 533)	=	4398.36
Residual	.674274319	533	.001265055	Prob > F	=	0.0000
				R-squared	=	0.9830
				Adj R-squared	=	0.9828
Total	39.6234122	540	.073376689	Root MSE	=	.03557
logDTCP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
mon	-.0000244	.0000115	-2.12	0.034	-.000047	-1.81e-06
logDTCP						
L1.	1.221972	.047602	25.67	0.000	1.128461	1.315482
L2.	-.2148846	.0734647	-2.93	0.004	-.3592004	-.0705688
L3.	-.0405285	.0475008	-0.85	0.394	-.1338402	.0527833
logEX						
L1.	-.1284591	.1215852	-1.06	0.291	-.3673041	.1103858
L2.	-.0413113	.1884555	-0.22	0.827	-.411518	.3288954
L3.	.1304735	.1210537	1.08	0.282	-.1073274	.3682744
_cons	.0454484	.014387	3.16	0.002	.0171862	.0737106

Appendix 17: Granger-causality Test – first difference

. regress d.logDTCP mon L(1/2).d.logDTCP L(1/2).d.logEX						
Source	SS	df	MS	Number of obs	=	541
				F(5, 535)	=	10.10
Model	.065022151	5	.01300443	Prob > F	=	0.0000
Residual	.688543229	535	.001286997	R-squared	=	0.0863
				Adj R-squared	=	0.0777
Total	.753565379	540	.001395491	Root MSE	=	.03587
D.logDTCP	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
mon	-4.26e-06	9.89e-06	-0.43	0.667	-.0000237	.0000152
logDTCP						
LD.	.2457558	.0474344	5.18	0.000	.1525752	.3389363
L2D.	.0215443	.0475699	0.45	0.651	-.0719024	.114991
logEX						
LD.	-.1064613	.1219414	-0.87	0.383	-.346004	.1330814
L2D.	-.1502968	.1217506	-1.23	0.218	-.3894645	.088871
_cons	.0014163	.0044061	0.32	0.748	-.0072391	.0100717

Appendix 19: VAR for Granger-causality Test

```
. vargranger
```

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2
D_logDTCP	D.logEX	2.9195	2	0.232
D_logDTCP	ALL	2.9195	2	0.232
D_logEX	D.logDTCP	5.5581	2	0.062
D_logEX	ALL	5.5581	2	0.062

Appendix 20: Cointegration test for log commodity price Agriculture

```
. varsoc ln_Agric, exo(mon)
```

Selection-order criteria

Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-3.08316				.059663	.018827	.025043	.034721
1	1054.83	2115.8	1	0.000	.00119	-3.89568	-3.88635	-3.87184
2	1090.87	72.082*	1	0.000	.001045	-4.02546	-4.01303*	-3.99367*
3	1092.01	2.2747	1	0.132	.001045	-4.02597	-4.01043	-3.98623
4	1093.92	3.8123	1	0.051	.001041*	-4.02932*	-4.01068	-3.98164

Endogenous: ln_Agric

Exogenous: mon _cons

```
. dfuller ln_Agric, lag (4) trend
```

Augmented Dickey-Fuller test for unit root

Number of obs = 539

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)	-1.986	-3.960	-3.410	-3.120

MacKinnon approximate p-value for Z(t) = 0.6091

```
. dfuller d.Agric , lag (4) trend //shows agriculture is I(1)
```

Augmented Dickey-Fuller test for unit root

Number of obs = 538

Test Statistic	Interpolated Dickey-Fuller			
	1% Critical Value	5% Critical Value	10% Critical Value	
Z(t)	-9.796	-3.960	-3.410	-3.120

MacKinnon approximate p-value for Z(t) = 0.0000

Appendix 21: Cointegration test for log commodity price Metals & Minerals

```
. varsoc ln_MetalsandMinerals, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4 Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-58.6678				.073301	.224696	.230912	.24059
1	1040.77	2198.9	1	0.000	.001254	-3.8436	-3.83427	-3.81976
2	1062.29	43.044*	1	0.000	.001162*	-3.9196*	-3.90717*	-3.88782*
3	1062.42	.25772	1	0.612	.001166	-3.91638	-3.90084	-3.87664
4	1063.03	1.217	1	0.270	.001168	-3.91493	-3.89628	-3.86724

Endogenous: ln_MetalsandMinerals
Exogenous: mon _cons

```
. dfuller ln_MetalsandMinerals , lag (3) trend
```

Augmented Dickey-Fuller test for unit root Number of obs = 540

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-2.137	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.5251

```
. dfuller d.ln_MetalsandMinerals , lag (3) trend //shows energy is I(1)
```

Augmented Dickey-Fuller test for unit root Number of obs = 539

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-10.537	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0000

Appendix 22: Cointegration test for log commodity price Energy

```
. varsoc ln_Energy, exo(mon)
```

Selection-order criteria

Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-284.07				.168915	1.05952	1.06573	1.07541
1	697.24	1962.6	1	0.000	.004475	-2.57126	-2.56194	-2.54742
2	717.948	41.414*	1	0.000	.00416*	-2.64425*	-2.63182*	-2.61246*
3	717.955	.01539	1	0.901	.004176	-2.64058	-2.62503	-2.60084
4	718.272	.63375	1	0.426	.004186	-2.63805	-2.6194	-2.59036

Endogenous: ln_Energy

Exogenous: mon _cons

```
. dfuller ln_Energy, lag (2) trend
```

Augmented Dickey-Fuller test for unit root Number of obs = 541

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-2.644	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.2602

```
. dfuller d.Energy , lag (2) trend //shows agriculture is I(1)
```

Augmented Dickey-Fuller test for unit root Number of obs = 540

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-11.763	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0000

Appendix 23: Cointegration test for log forest

```
. varsoc ln_Forest, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4 Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	360.492				.01552	-1.32775	-1.32153	-1.31185
1	1024.46	1327.9	1	0.000	.001332	-3.78318	-3.77385	-3.75934
2	1032.07	15.227*	1	0.000	.0013*	-3.80767*	-3.79524*	-3.77588*
3	1032.56	.97357	1	0.324	.001302	-3.80577	-3.79023	-3.76603
4	1032.56	.00402	1	0.949	.001307	-3.80207	-3.78342	-3.75439

Endogenous: ln_Forest
Exogenous: mon _cons

```
. dfuller ln_Forest , lag (2) trend
```

Augmented Dickey-Fuller test for unit root Number of obs = 541

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-3.581	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0314

```
. dfuller d.ln_Forest, lag (2) trend //shows forestry is I(1)
```

Augmented Dickey-Fuller test for unit root Number of obs = 540

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-13.458	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0000

Appendix 24: Cointegration test for log commodity price Fish

```
. varsoc ln_Fish, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4

Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	72.5614				.045085	-.261339	-.255122	-.245444
1	376.998	608.87	1	0.000	.014654	-1.38518	-1.37585	-1.36134
2	380.363	6.7306	1	0.009	.014526	-1.39394	-1.38151	-1.36215
3	384.267	7.8077*	1	0.005	.014371*	-1.40469*	-1.38915*	-1.36496*
4	384.268	.00132	1	0.971	.014424	-1.40099	-1.38234	-1.35331

Endogenous: ln_Fish
Exogenous: mon _cons

```
. dfuller ln_Fish , lag (3) trend
```

Augmented Dickey-Fuller test for unit root

Number of obs = 540

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-6.456	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0000

```
. dfuller d.ln_Fish , lag (3) trend //shows fish is I(0)
```

Augmented Dickey-Fuller test for unit root

Number of obs = 539

Test Statistic	Interpolated Dickey-Fuller		
	1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-14.108	-3.960	-3.410

MacKinnon approximate p-value for Z(t) = 0.0000

Appendix 25: Cointegration test for log of commodity prices for Agriculture and Metals & Minerals

```

. varsoc ln_Agric ln_MetalsandMinerals, exo(mon)

```

Selection-order criteria								
Sample: 1972m5 - 2017m4					Number of obs		=	540
lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	179.927				.001787	-.651581	-.639148	-.619792
1	2105.42	3851	4	0.000	1.4e-06	-7.76823	-7.74337	-7.70465
2	2163.75	116.66*	4	0.000	1.2e-06*	-7.96945*	-7.93215*	-7.87408*
3	2167	6.4999	4	0.165	1.2e-06	-7.96667	-7.91694	-7.83952
4	2169.72	5.4324	4	0.246	1.2e-06	-7.96192	-7.89975	-7.80297

```

Endogenous: ln_Agric ln_MetalsandMinerals
Exogenous: mon _cons

```

```

. vecrank ln_Agric ln_MetalsandMinerals, trend(trend) lags(2) //no cointegration

```

Johansen tests for cointegration

Trend: trend		Number of obs =		542	
Sample: 1972m3 - 2017m4		Lags =		2	
maximum rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	8	2163.2271	.	20.7931	18.17
1	11	2171.8759	0.03141	3.4955*	3.74
2	12	2173.6236	0.00643		

```

. var d_ln_Agric d_ln_MetalsandMinerals, lags(1/3) exog(mon)

```

Vector autoregression

Sample: 1972m5 - 2017m4		Number of obs		=	540
Log likelihood = 2161.145		AIC		=	-7.944982
FPE = 1.22e-06		HQIC		=	-7.895251
Det(Sigma_ml) = 1.15e-06		SBIC		=	-7.817824
Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_ln_Agric	8	.032126	0.1404	88.23462	0.0000
D_ln_Metalsand~s	8	.034054	0.0827	48.66477	0.0000

		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D_ln_Agric							
	ln_Agric						
	LD.	.3564878	.0431134	8.27	0.000	.2719872	.4409884
	L2D.	-.0485468	.0456315	-1.06	0.287	-.1379829	.0408894
	L3D.	-.101204	.043169	-2.34	0.019	-.1858136	-.0165943
ln_MetalsandMinerals							
	LD.	.0341509	.0408764	0.84	0.403	-.0459653	.1142671
	L2D.	.0691887	.0422312	1.64	0.101	-.013583	.1519603
	L3D.	.0098133	.0411283	0.24	0.811	-.0707966	.0904232
	mon	1.81e-06	8.80e-06	0.21	0.837	-.0000154	.0000191
	_cons	-.0022442	.0039265	-0.57	0.568	-.0099399	.0054516
D_ln_MetalsandMinerals							
	ln_Agric						
	LD.	.0372744	.0457004	0.82	0.415	-.0522967	.1268455
	L2D.	.0674498	.0483696	1.39	0.163	-.027353	.1622526
	L3D.	.0080643	.0457593	0.18	0.860	-.0816223	.0977509
ln_MetalsandMinerals							
	LD.	.257462	.0433291	5.94	0.000	.1725385	.3423856
	L2D.	-.0056966	.0447653	-0.13	0.899	-.093435	.0820418
	L3D.	.0320773	.0435962	0.74	0.462	-.0533696	.1175242
	mon	-3.25e-07	9.33e-06	-0.03	0.972	-.0000186	.000018
	_cons	.0001906	.0041621	0.05	0.963	-.0079669	.0083482

Appendix 26: Cointegration test for log of commodity prices for Agriculture and Metals & Minerals

```
. varstable, graph
```

Eigenvalue stability condition

Eigenvalue	Modulus
.3294169 + .4184189i	.532532
.3294169 - .4184189i	.532532
.4662241	.466224
-.3480319	.348032
-.08153808 + .2561649i	.268829
-.08153808 - .2561649i	.268829

All the eigenvalues lie inside the unit circle.
VAR satisfies stability condition.

```
. varnorm, jbera skewness kurtosis
```

Jarque-Bera test

Equation	chi2	df	Prob > chi2
D_ln_Agric	178.580	2	0.00000
D_ln_MetalsandMinerals	105.237	2	0.00000
ALL	283.817	4	0.00000

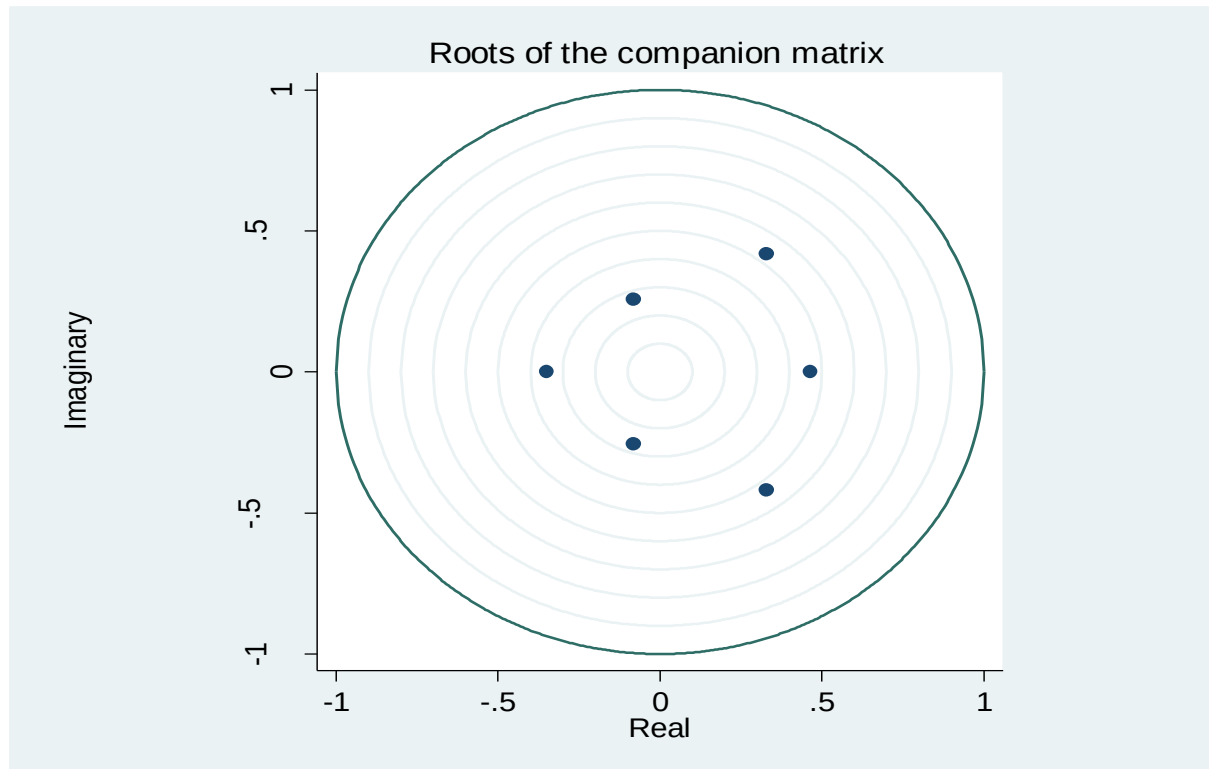
Skewness test

Equation	Skewness	chi2	df	Prob > chi2
D_ln_Agric	.1105	1.099	1	0.29452
D_ln_MetalsandMinerals	-.14978	2.019	1	0.15533
ALL		3.118	2	0.21035

Kurtosis test

Equation	Kurtosis	chi2	df	Prob > chi2
D_ln_Agric	5.8086	177.481	1	0.00000
D_ln_MetalsandMinerals	5.1418	103.218	1	0.00000
ALL		280.699	2	0.00000

Unit Circle for log of commodity prices for Agriculture and Metals & Minerals



Appendix 27: Cointegration test for log of commodity prices for Metals & Minerals and Energy

```
. varsoc ln_MetalsandMinerals ln_Energy, exo(mon)
```

Selection-order criteria

Sample: 1972m5 - 2017m4

Number of obs

=

540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-234.544				.008294	.883496	.895929	.915286
1	1751.72	3972.5	4	0.000	5.4e-06	-6.45823	-6.43336	-6.39465
2	1792.31	81.186*	4	0.000	4.7e-06*	-6.59376*	-6.55646*	-6.49839*
3	1793.71	2.7978	4	0.592	4.7e-06	-6.58412	-6.53439	-6.45697
4	1795.17	2.9065	4	0.574	4.8e-06	-6.57469	-6.51253	-6.41574

Endogenous: ln_MetalsandMinerals ln_Energy

Exogenous: mon _cons

```
. vecrank ln_MetalsandMinerals ln_Energy, trend(trend) lags(2) //no cointegration
```

Johansen tests for cointegration

Trend: trend

Number of obs =

542

Sample: 1972m3 - 2017m4

Lags =

2

maximum rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	8	1793.1709	.	15.3389*	18.17
1	11	1798.6061	0.01986	4.4685	3.74
2	12	1800.8404	0.00821		

```
. var d.ln_MetalsandMinerals d.ln_Energy, lags(1/3) exo(mon)
```

Vector autoregression

Sample: 1972m5 - 2017m4

Number of obs

=

540

Log likelihood = 1788.403

AIC

=

-6.564455

FPE = 4.83e-06

HQIC

=

-6.514724

Det(Sigma_ml) = 4.55e-06

SBIC

=

-6.437297

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_ln_Metalsand~s	8	.034086	0.0809	47.54524	0.0000
D_ln_Energy	8	.06447	0.0837	49.31215	0.0000

	Coef.	Std. Err.	z	P> z	[95% Conf. Interv	
D_ln_MetalsandMinerals						
ln_MetalsandMinerals						
LD.	.2565223	.0436469	5.88	0.000	.170976	.3420
L2D.	.0096408	.0451333	0.21	0.831	-.0788189	.0981
L3D.	.0415441	.0440035	0.94	0.345	-.0447012	.1277
ln_Energy						
LD.	.0357633	.0230472	1.55	0.121	-.0094085	.0809
L2D.	-.0229451	.0237404	-0.97	0.334	-.0694754	.0235
L3D.	-.0124884	.0230418	-0.54	0.588	-.0576495	.0326
mon	5.69e-09	9.36e-06	0.00	1.000	-.0000183	.0000
_cons	-.0001514	.0041704	-0.04	0.971	-.0083252	.0080
D_ln_Energy						
ln_MetalsandMinerals						
LD.	.0660749	.0825528	0.80	0.423	-.0957256	.2278
L2D.	.1142935	.0853642	1.34	0.181	-.0530173	.2816
L3D.	.1154134	.0832274	1.39	0.166	-.0477093	.2785
ln_Energy						
LD.	.2499108	.0435911	5.73	0.000	.1644738	.3353
L2D.	-.0091823	.0449021	-0.20	0.838	-.0971888	.0788
L3D.	-.0596314	.0435808	-1.37	0.171	-.1450481	.0257
mon	-.0000128	.0000177	-0.72	0.470	-.0000475	.0000
_cons	.0061846	.0078878	0.78	0.433	-.0092752	.0216

Appendix 27: Cointegration test for log of commodity prices for Metals & Minerals and Energy

```
. varstable, graph
```

Eigenvalue stability condition

Eigenvalue	Modulus
.2869505 + .2200841i	.361632
.2869505 - .2200841i	.361632
.3342929	.334293
-.3129109	.312911
-.04442503 + .2715853i	.275195
-.04442503 - .2715853i	.275195

All the eigenvalues lie inside the unit circle.
VAR satisfies stability condition.

```
. varnorm, jbera skewness kurtosis
```

Jarque-Bera test

Equation	chi2	df	Prob > chi2
D_ln_MetalsandMinerals	110.579	2	0.00000
D_ln_Energy	56.162	2	0.00000
ALL	166.741	4	0.00000

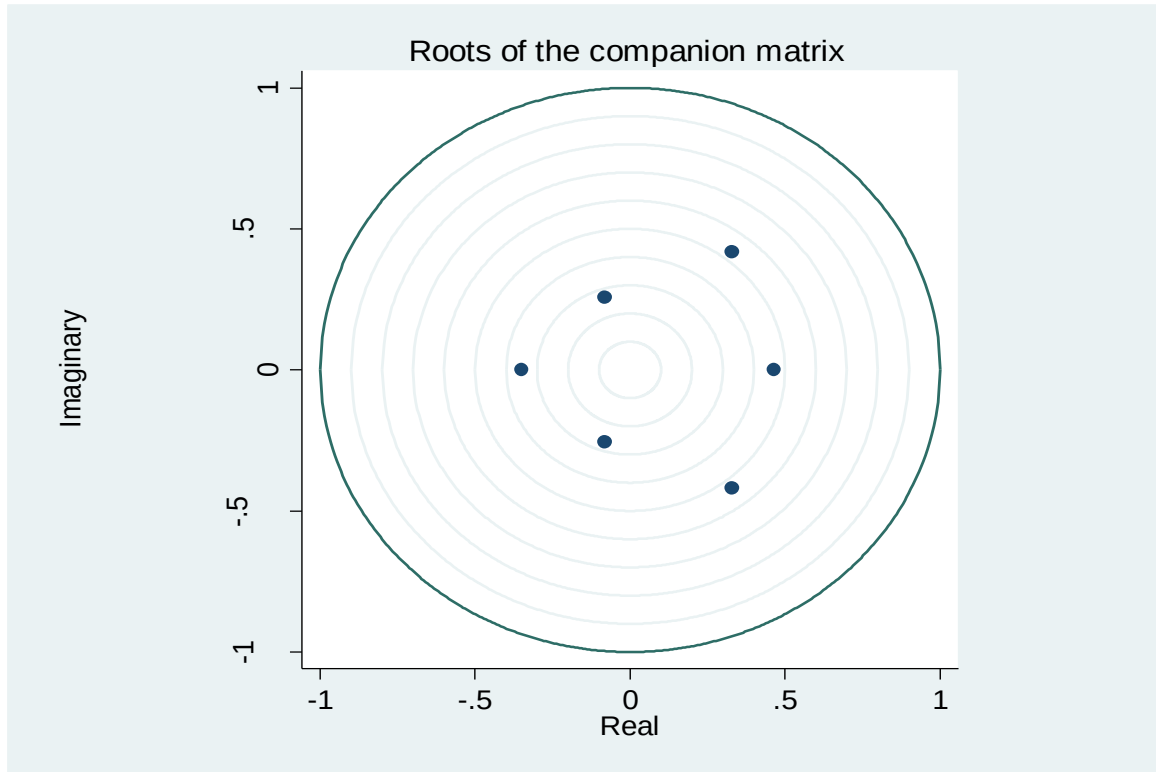
Skewness test

Equation	Skewness	chi2	df	Prob > chi2
D_ln_MetalsandMinerals	-.20271	3.698	1	0.05447
D_ln_Energy	-.11634	1.218	1	0.26972
ALL		4.916	2	0.08559

Kurtosis test

Equation	Kurtosis	chi2	df	Prob > chi2
D_ln_MetalsandMinerals	5.17951	106.881	1	0.00000
D_ln_Energy	4.5627	54.944	1	0.00000
ALL		161.824	2	0.00000

Unit Circle for log of commodity prices for Agriculture and Metals & Minerals.



Appendix 28: Cointegration test for log of commodity prices for Agriculture, Metals & Minerals, Energy and Forest.

```
. varsoc ln_Agric ln_Energy ln_MetalsandMinerals ln_Forest, exo(mon)
```

Selection-order criteria
Sample: 1972m5 - 2017m4 Number of obs = 540

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	429.852				2.5e-06	-1.56241	-1.53755	-1.49884
1	3854.05	6848.4	16	0.000	8.1e-12	-14.1854	-14.1108	-13.9946
2	3941.09	174.06*	16	0.000	6.2e-12*	-14.4485*	-14.3241*	-14.1306*
3	3948.78	15.391	16	0.496	6.4e-12	-14.4177	-14.2437	-13.9727
4	3955.49	13.427	16	0.641	6.7e-12	-14.3833	-14.1595	-13.8111

Endogenous: ln_Agric ln_Energy ln_MetalsandMinerals ln_Forest
Exogenous: mon _cons

```
. vecrank ln_Agric ln_Energy ln_MetalsandMinerals ln_Forest, trend(trend) lags(2)
```

Johansen tests for cointegration
Trend: trend Number of obs = 542
Sample: 1972m3 - 2017m4 Lags = 2

rank	parms	LL	eigenvalue	trace statistic	5% critical value
0	24	3931.7694	.	55.1983	54.64
1	31	3943.256	0.04150	32.2251*	34.55
2	36	3952.9166	0.03502	12.9039	18.17
3	39	3957.4787	0.01669	3.7798	3.74
4	40	3959.3686	0.00695		

```
. var d.ln_Agric d.ln_Energy d.ln_MetalsandMinerals d.ln_Forest, lags(1/3) exo(mon)
```

Vector autoregression
Sample: 1972m5 - 2017m4 Number of obs = 540
Log likelihood = 3931.009 AIC = -14.35188
FPE = 6.87e-12 HQIC = -14.17783
Det(Sigma_ml) = 5.59e-12 SBIC = -13.90683

Equation	Parms	RMSE	R-sq	chi2	P>chi2
D_ln_Agric	14	.032206	0.1459	92.26014	0.0000
D_ln_Energy	14	.064532	0.0923	54.88064	0.0000
D_ln_Metalsand~s	14	.034123	0.0893	52.95111	0.0000
D_ln_Forest	14	.03623	0.0478	27.12971	0.0119

		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
D_ln_Agric	ln_Agric						
	LD.	.3569102	.0432364	8.25	0.000	.2721684	.4416521
	L2D.	-.0466621	.045768	-1.02	0.308	-.1363657	.0430416
	L3D.	-.1082556	.0435002	-2.49	0.013	-.1935144	-.0229968
	ln_Energy						
	LD.	-.0030146	.0217729	-0.14	0.890	-.0456887	.0396595
	L2D.	-.0030719	.0223763	-0.14	0.891	-.0469287	.0407849
	L3D.	.0019723	.0216857	0.09	0.928	-.0405309	.0444756
	ln_MetalsandMinerals						
	LD.	.0393364	.0415559	0.95	0.344	-.0421118	.1207845
	L2D.	.068285	.0428734	1.59	0.111	-.0157453	.1523153
	L3D.	.0029	.0420024	0.07	0.945	-.0794233	.0852233
	ln_Forest						
	LD.	-.0130038	.0387523	-0.34	0.737	-.088957	.0629494
	L2D.	.0061274	.0389959	0.16	0.875	-.0703031	.0825579
	L3D.	.067551	.0389332	1.74	0.083	-.0087567	.1438588
	mon	1.80e-06	8.79e-06	0.20	0.838	-.0000154	.000019
	_cons	-.0021842	.0039227	-0.56	0.578	-.0098725	.0055042
D_ln_Energy	ln_Agric						
	LD.	.1088403	.0866352	1.26	0.209	-.0609616	.2786423
	L2D.	.0402086	.0917079	0.44	0.661	-.1395355	.2199528
	L3D.	-.0352731	.0871637	-0.40	0.686	-.2061109	.1355647
	ln_Energy						
	LD.	.2470457	.0436276	5.66	0.000	.1615373	.3325542
	L2D.	-.0080813	.0448367	-0.18	0.857	-.0959596	.079797
	L3D.	-.0577585	.0434529	-1.33	0.184	-.1429246	.0274077
	ln_MetalsandMinerals						
	LD.	.0628252	.0832679	0.75	0.451	-.100377	.2260273
	L2D.	.1113248	.0859078	1.30	0.195	-.0570514	.2797009
	L3D.	.10334	.0841626	1.23	0.219	-.0616157	.2682957
	ln_Forest						
	LD.	-.1109392	.0776502	-1.43	0.153	-.2631308	.0412524
	L2D.	.0849607	.0781382	1.09	0.277	-.0681873	.2381088
	L3D.	-.0109067	.0780127	-0.14	0.889	-.1638088	.1419953
	mon	-.0000131	.0000176	-0.74	0.457	-.0000476	.0000214
	_cons	.0064986	.0078601	0.83	0.408	-.008907	.0219041
D_ln_MetalsandMinerals	ln_Agric						
	LD.	.0309833	.0458109	0.68	0.499	-.0588045	.1207711
	L2D.	.0705418	.0484933	1.45	0.146	-.0245033	.1655868
	L3D.	.0096062	.0460904	0.21	0.835	-.0807294	.0999417
	ln_Energy						
	LD.	.0314823	.0230694	1.36	0.172	-.0137329	.0766974
	L2D.	-.0247791	.0237087	-1.05	0.296	-.0712474	.0216891
	L3D.	-.0121435	.022977	-0.53	0.597	-.0571776	.0328907
	ln_MetalsandMinerals						
	LD.	.2486114	.0440304	5.65	0.000	.1623134	.3349093
	L2D.	.0026516	.0454263	0.06	0.953	-.0863823	.0916854
	L3D.	.0371383	.0445035	0.83	0.404	-.0500869	.1243635
	ln_Forest						
	LD.	.0148276	.0410598	0.36	0.718	-.0656482	.0953033
	L2D.	-.0353063	.0413179	-0.85	0.393	-.1162878	.0456753
	L3D.	.0110501	.0412515	0.27	0.789	-.0698014	.0919015
	mon	-3.23e-07	9.32e-06	-0.03	0.972	-.0000186	.0000179
	_cons	.0001913	.0041563	0.05	0.963	-.0079549	.0083374

D_ln_Forest							
ln_Agric							
LD.	.0303979	.0486396	0.62	0.532	-.064934	.1257298	
L2D.	.0694186	.0514875	1.35	0.178	-.0314951	.1703323	
L3D.	.0163432	.0489363	0.33	0.738	-.0795702	.1122567	
ln_Energy							
LD.	.0001612	.0244938	0.01	0.995	-.0478459	.0481682	
L2D.	-.0061839	.0251727	-0.25	0.806	-.0555214	.0431536	
L3D.	.0182642	.0243958	0.75	0.454	-.0295507	.066079	
ln_MetalsandMinerals							
LD.	.1011404	.0467491	2.16	0.031	.0095139	.192767	
L2D.	.0202771	.0482312	0.42	0.674	-.0742543	.1148085	
L3D.	.014343	.0472514	0.30	0.761	-.078268	.1069541	
ln_Forest							
LD.	.1268525	.0435951	2.91	0.004	.0414076	.2122974	
L2D.	-.0827851	.0438691	-1.89	0.059	-.168767	.0031968	
L3D.	-.0278259	.0437986	-0.64	0.525	-.1136697	.0580179	
mon	2.90e-06	9.89e-06	0.29	0.769	-.0000165	.0000223	
_cons	-.0015996	.0044129	-0.36	0.717	-.0102487	.0070495	

. varstable, graph

Eigenvalue stability condition

Eigenvalue	Modulus
.3231838 + .46522i	.56646
.3231838 - .46522i	.56646
.4455346	.445535
.3314712 + .251148i	.415871
.3314712 - .251148i	.415871
.150996 + .344982i	.37658
.150996 - .344982i	.37658
-.3710073	.371007
-.2846036 + .1060824i	.303731
-.2846036 - .1060824i	.303731
-.06860122 + .2808787i	.289135
-.06860122 - .2808787i	.289135

All the eigenvalues lie inside the unit circle.
VAR satisfies stability condition.

. varnorm, jbera skewness kurtosis

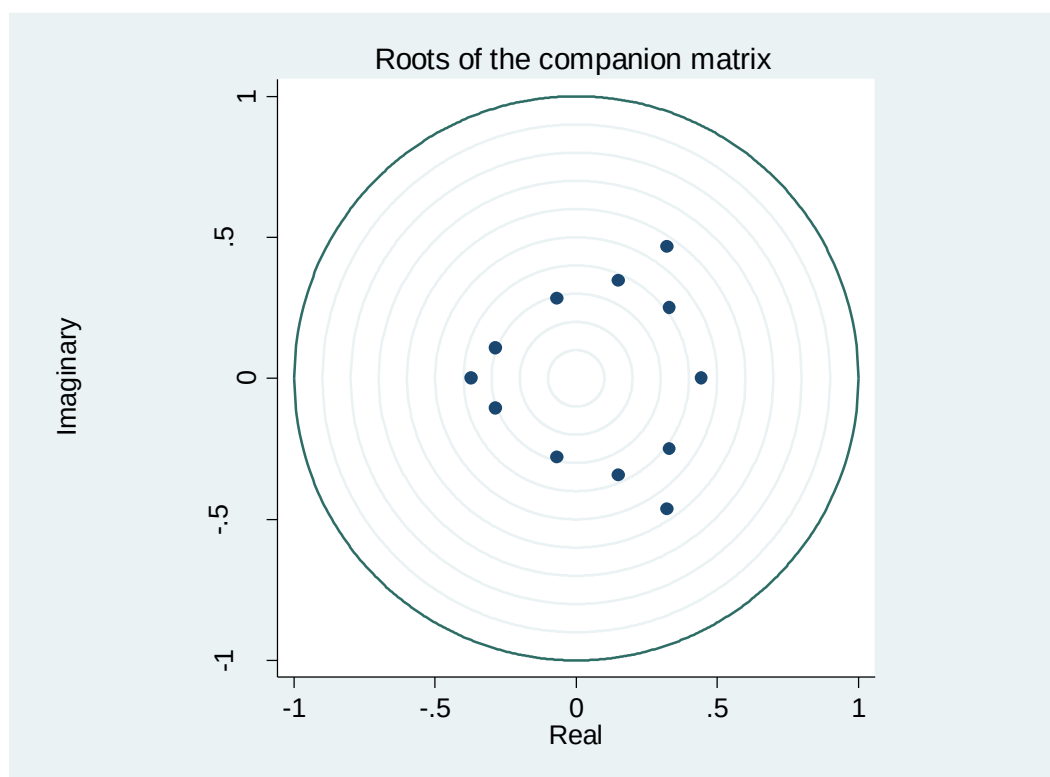
Jarque-Bera test

Equation	chi2	df	Prob > chi2
D_ln_Agric	175.344	2	0.00000
D_ln_Energy	70.927	2	0.00000
D_ln_MetalsandMinerals	99.812	2	0.00000
D_ln_Forest	46.085	2	0.00000
ALL	392.169	8	0.00000

Skewness test					
Equation	Skewness	chi2	df	Prob > chi2	
D_ln_Agric	.10626	1.016	1	0.31343	
D_ln_Energy	-.07218	0.469	1	0.49352	
D_ln_MetalsandMinerals	-.15848	2.260	1	0.13272	
D_ln_Forest	.07131	0.458	1	0.49873	
ALL		4.203	4	0.37922	

Kurtosis test					
Equation	Kurtosis	chi2	df	Prob > chi2	
D_ln_Agric	5.7835	174.328	1	0.00000	
D_ln_Energy	4.7696	70.458	1	0.00000	
D_ln_MetalsandMinerals	5.0822	97.552	1	0.00000	
D_ln_Forest	4.424	45.628	1	0.00000	
ALL		387.965	4	0.00000	

Unit Circle for log of commodity prices for Agriculture, Metals & Minerals, Energy, and Forestry.



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